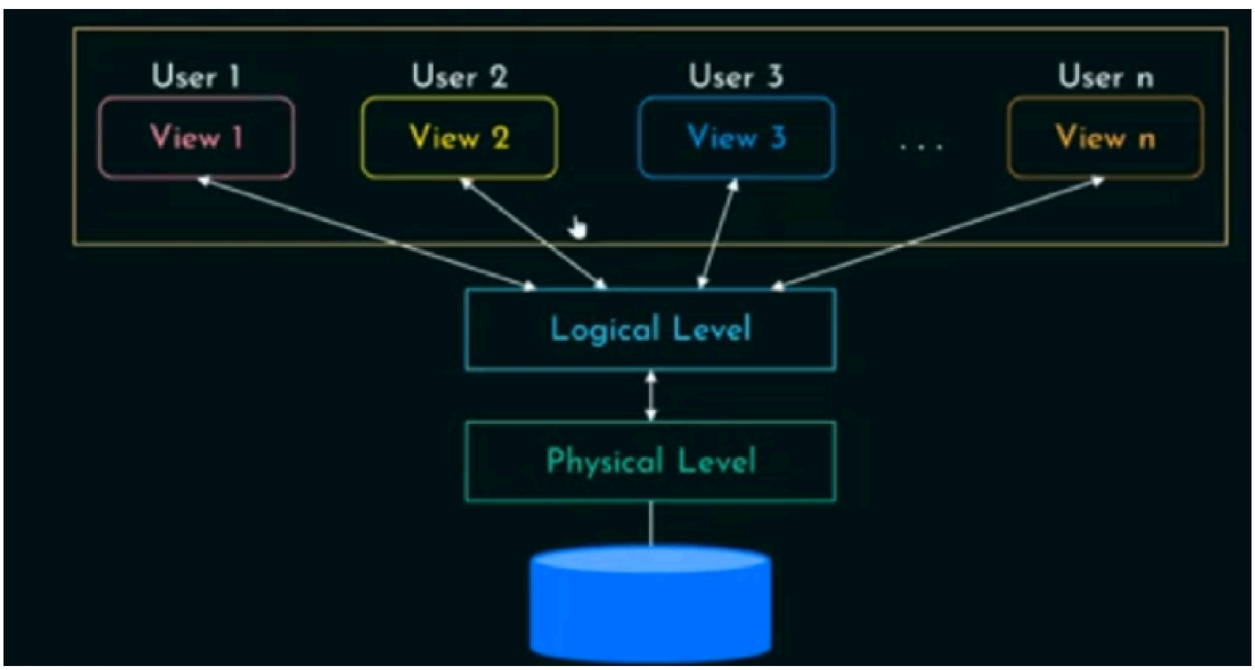
PURPOSE OF DATABASE SYSTEM

- Reduces data redundancy
- Ensures data consistency
- Allows concurrent access by multiple users
- Maintains data integrity
- Provides data security
- Enables easier data access and sharing



What is Data Abstraction? (5 Simple Points)

- **Data abstraction means hiding unnecessary details and showing only the needed information.**

- **It hides complex information**

- **A database stores a lot of data, but users don't need to see all of it**

- **Users don't need to know how data is stored or managed, they just access what they need.**

- **This makes databases easier to use and understand.**
- **There are three levels of data abstraction: Physical, Logical, and View.**
- **Example: A library has thousands of books, but you only search for the one you need.**

Physical Level (Lowest Level) – How Data is Stored

- **Describes how and where data is saved in memory.**
- **Lowest level of abstraction**

- **Includes files, indexing, and data storage techniques.**

- **Users do not see this level; only database administrators manage it.**

- **Example: Data is stored in hard disks, SSDs, or cloud servers.**

Logical Level (Middle Level) – Key Points:

1. **Describes what data is stored and how it is related.**
2. **Uses tables, columns, and relationships to organize data.**
3. **Focuses on the structure of the database rather than storage details**
4. **Supports logical data independence — changes in storage do not affect the logical structure.**
5. **Example: A Student table with fields like *Name*, *ID*, and *Course*.**

3. View Level (Highest Level) – User Interaction

- **Shows only the necessary data to specific users.**
- **Hides complexity and protects sensitive information.**
- **Users interact with this level using applications or queries.**
- **Provides different views for different users (e.g., a teacher sees grades, a student sees personal details).**
- **Example: A Bank App where customers see only their account details, not the bank's full database**

Introduction to Relational Databases (Simplified)

1. **A relational database stores data in tables (also called relations).**
2. **Each table has columns (attributes) and rows (tuples).**

3. Each row in a table shows details about one item — like one instructor.

4.Each column stores one type of data, like numbers, names, or dates.

Keys in DBMS

A key is something that helps you find one specific row in a table.

It helps to find specific records easily and quickly.

Keys are used to connect tables with each other.

A key value is unique — no two rows have the same key.

SUPER KEY:

https://youtu.be/HtcYgIR1Y_k?si=Usm42bsz6AoQ9pvR

PRIMARY KEY:

COLUMN NAME	UNIQUE VALUES	NO NULL VALUES	PRIMARY KEY
NAME	✗	✓	
AGE	✗	✓	

https://youtu.be/B2zInY06CaE?si=JZreGxrdiXYUxatb

CANDIDATE KEY:

Alternate key:

candidate key has primary key there will be only one primary key in the table in candidate key other than primary key all other are alternate key say these three sentences are crt or not

https://youtu.be/uau8R_WhAI0?si=Dj888a6R7LGhvGy-

FORIEGN KEY:

To link both tables: